

Equations from Quantities



Grade 7 • Mathematics
NC.7.EE.4

Model and solve multi-step equations.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

The school garden pays a \$2 setup fee plus \$4 for each kit. The total is \$78. Write and solve an equation for the number of kits.

Let x count kits. $4x + 2 = 78$; $4x = 76$; $x = 19$. Check: $4(19) + 2 = 78$.



Build a second model together

The reading group pays a \$5 setup fee plus \$2 for each kit. The total is \$27. Write and solve an equation for the number of kits.

Discuss: What should the first step find?



Your turn • 1

The reading group pays a \$6 setup fee plus \$6 for each kit. The total is \$84. Write and solve an equation for the number of kits.

Represent → solve → explain → check



Your turn • 2

The library team pays a \$3 setup fee plus \$6 for each kit. The total is \$39. Write and solve an equation for the number of kits.

Represent → solve → explain → check



Your turn • 3

The design team pays a \$8 setup fee plus \$4 for each kit. The total is \$84. Write and solve an equation for the number of kits.

Represent → solve → explain → check



Your turn • 4

The community center pays a \$5 setup fee plus \$8 for each kit. The total is \$125. Write and solve an equation for the number of kits.

Represent → solve → explain → check



Exit question • Explain independently

The science club pays a \$2 setup fee plus \$4 for each kit. The total is \$14. Write and solve an equation for the number of kits.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 13 kits

2. 6 kits

3. 19 kits

4. 15 kits

Exit: 3 kits

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

